Paper A-02: Earth Atmospheric Circulation: A Constraint-Driven Flux Dynamics (CDFD) Perspective

Steve Bico Mujjabi, MD
Independent Researcher
ORCID: 0009-0001-0556-5516

May 2026

Abstract

This paper treats atmospheric circulation as a CDFD/AFL modelling hypothesis. Solar heating, latent heat, water vapour, aerosols, and momentum are read as driving fluxes Φ moving through pressure gradients, rotation, stratified layers, topography, and boundary-layer drag. The purpose is not to replace meteorology or climate dynamics. The purpose is to name the measurable flux, constraint, responsiveness, and memory terms that make atmospheric overload or stagnation testable.

1 Series Context

Part I sets the flow–constraint language used here [1]. Part II develops the Life Number and the origins-of-life use of the same notation [2]. Part III applies AFL to biology and medicine [3]. Part IV asks whether the same audit language clarifies cross-domain measurement without turning analogy into evidence.

Notation. In this paper, C names the active atmospheric constraint or capacity burden. Φ is the driving flow, S is the surface or circulation response, and M_s is retained structural memory. Λ appears only as the Part II Life Number reference [2]; it is not an atmospheric variable in this manuscript.

2 Atmospheric Transport Frame

The atmosphere is a moving constraint network. Uneven heating creates pressure gradients; rotation bends flow; mountains, land–sea contrast, cloud formation, surface roughness, and radiative forcing reshape the pathways available to air and moisture. CDFD/AFL treats those pathways as an audit surface: where does flux rise, where does the atmosphere lose room to adjust, and where does memory from prior heat, moisture, or aerosol loading still shape the next state?

CDFD term	Atmospheric reading	Example proxy
Φ	heat, moisture, aerosol, or momentum flux	radiative flux, latent heat, wind stress
C	active circulation constraint	inversion strength, topography, drag, stab
S	responsiveness	convective adjustment, mixing, storm-trac
M_s	structural memory	soil moisture, sea-surface heat, aerosol per
Ψ_s	operating ratio	ventilated, stagnant, amplified, or overloa

3 Candidate Variable Map

4 Overload and Stagnation

An atmospheric overload claim is meaningful only when the flux and constraint are named together. A heat-wave example might use radiative and sensible heat flux as Φ, boundary-layer stability and weak advection as C, convective mixing as S, and stored soil or ocean heat as M_s. A pollution-stagnation example might use emissions as Φ, inversion strength as C, turbulent mixing as S, and accumulated aerosol load as M_s. These mappings remain hypotheses until compared with atmospheric observations and standard models.

5 Measurement Layer

The measurable layer comes first. A useful atmospheric application gives Φ, C, S, and M_s observable proxies, a spatial grid or network, a time window, and a failure condition such as persistent stagnation, blocked moisture transport, convective inhibition, or extreme heat amplification. If conventional fluid-dynamical variables explain the event without added value from CDFD/AFL, the mapping adds no release-level claim.

6 Current Runtime Diagnostic

The Part E diagnostic run includes a climate/atmosphere adapter row and the universal cascade stress test. These files check finite runtime behavior and regime reporting in the current CDFD Runtime [4]. They are a model audit for later measurement, not a replacement for atmospheric data.

7 Tests That Can Break the Mapping

The atmospheric mapping weakens if measured flux and constraint proxies do not track stagnation, amplification, or recovery; if the same event is explained more cleanly by established atmospheric diagnostics; or if the CDFD/AFL terms cannot be separated from ordinary variable names in a reproducible analysis.

8 Conclusion

Atmospheric circulation can be read as constrained transport when the variables are tied to observations. The CDFD/AFL contribution is a common audit language for flux, bottleneck, response, and memory. Its value depends on whether that language sharpens measurable atmospheric questions.

Conflict of Interest

The author declares no conflict of interest.

Data Availability

Release-local runtime diagnostics are available under each Part's `outputs/` folder, with release-wide synthesis artifacts under `Part_E_Synthesis/supplementary/`, `Part_E_Synthesis/outputs` and `Part_E_Synthesis/figures/`.

9 Reference Layer

References

[1] Steve Bico Mujjabi. Cdfd part i: Fundamental physics - constraint-driven field theory and flux dynamics, 2026.

[2] Steve Bico Mujjabi. Cdfd part ii: Origins of life and tri-regime bioenergetics - constraint-driven flux dynamics, 2026.

[3] Steve Bico Mujjabi. Cdfd part iii: Adaptive flux limitation biology and medicine - constraint-driven flux dynamics, 2026.

[4] Steve Bico Mujjabi. Cdfd runtime: Constraint-driven flux dynamics and cdfl execution engine, 2026.